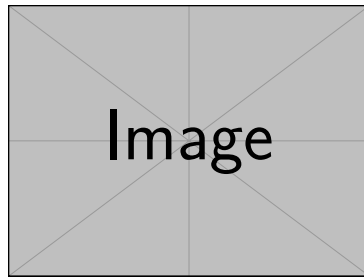


# Time Series Forecasting

Foundations and Practical Applications with Python



Yuvaraju

M.Tech - School of Computing, 2025

July 17, 2025



# M.Tech Syllabus Overview



*“Forecasting is not about predicting the future, it’s about reducing uncertainty.”* — Paul Saffo

## Module 1: Introduction to Time Series

- What is Time Series? Need for Forecasting
- Univariate and Multivariate Time Series
- Special Features of Time Series
- Exploratory Data Analysis (EDA)
- Components: Trend, Seasonality, Irregular Component
- Decomposition: Additive and Multiplicative Models
- Examples of Additive and Multiplicative Models

### Python Tasks:

- T1: EDA of time series signals
- T2: Decomposition analysis
- T3: Visualization of time series

## Module 2: Moving Average

- Simple Forecasting Methods
- Simple, Double, Triple Exponential Smoothing (Holt, Holt-Winters)
- Forecasting Examples

### Python Tasks:

- T4: Moving average model
- T5: Exponential smoothing
- T6: Model evaluation

## Module 3: Stationary Processes

- Stationary Processes and Tests (ADF)
- Lag, Differencing
- AR, MA, ARMA Processes
- ACF, PACF Analysis
- Parameter Estimation (Yule-Walker, MLE)

### Python Tasks:

- T7: Autoregression model
- T8: ARMA model
- T9: Case Study

## Module 4: Non-Stationary Processes

- ARIMA and SARIMA Models
- Akaike Information Criterion (AIC)
- Forecasting Examples

### Python Tasks:

- T10: ARIMA implementation
- T11: SARIMA modeling
- T12: Case Study

## Module 5: Multivariate Time Series

- VAR and VARMA Models
- Multivariate Forecasting
- Deep Learning Models for Time Series

### Python Tasks:

- T13: VAR implementation
- T14: VARMA modeling
- T15: Case Study

# Chapter 1

## Introduction to Time Series

### 1.1 What is a Time Series?

 “Forecasting is not about predicting the future, it’s about reducing uncertainty.” — Paul Saffo

A **time series** is a sequence of data points indexed in time order. These observations are typically spaced at uniform time intervals (e.g., daily, monthly, hourly). The temporal ordering of the data is critical in time series, unlike in traditional statistical datasets.

### Mini Story: Raju at Airtel – The Need for Time Series

#### Raju’s Forecasting Challenge at Airtel

Raju is a data analyst at Airtel. Every Monday morning, his Chief Growth Officer wants a report on how many new prepaid SIM activations are expected for the week across different regions. The activations from the past year show strong weekly patterns due to marketing campaigns and regional festivals.

Raju realizes that traditional linear regression can’t help here — the order of data, seasonality, and patterns matter too much. He turns to time series forecasting to analyze historical trends and build a model to predict future demand more accurately.

### Examples of Time Series Data:

- Daily stock prices of Infosys Ltd.
- Weekly prepaid SIM activations in Tamil Nadu (Raju’s dataset!)
- Monthly electricity usage of residential customers
- ECG signal readings from a wearable device over time
- Yearly rainfall levels in a river basin

### Key Idea

Time series analysis focuses on uncovering patterns like trend, seasonality, and irregular fluctuations to better understand the data and forecast future behavior.

## Sample Time Series Plot

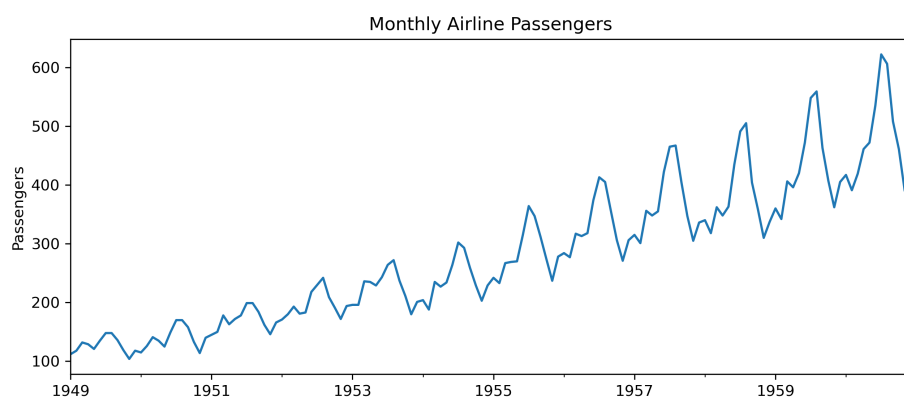


Figure 1.1: Monthly Airline Passengers Data (1949–1960) — Clear Trend and Seasonality

## 1.2 Need for Forecasting

*“The best way to predict the future is to understand the past.” — Abraham Lincoln*

Forecasting is the process of making predictions about the future based on historical and current data. In domains like finance, healthcare, supply chain, marketing, and energy, accurate forecasting enables better planning, resource allocation, and decision-making.

While traditional models often ignore time dependency, time series forecasting embraces the temporal nature of data. This allows analysts to capture underlying trends, seasonal effects, and autocorrelation, leading to more accurate predictions.

## Real-Life Story: Anita at BigBasket – The Inventory Dilemma

### Anita's Forecasting Dilemma at BigBasket

Anita is a Data Scientist at BigBasket, India's leading online grocery store. Every Friday evening, she must forecast how many packets of milk, eggs, and vegetables will be needed for each warehouse the following week.

If she overestimates demand, items spoil and lead to wastage. If she underestimates, customers find products out of stock, damaging brand loyalty. Previously, the company relied on simple average sales — but it ignored seasonality, festive spikes, and weekday patterns.

Anita proposes a time series forecasting model to analyze daily sales patterns from the past 18 months, factoring in trends and festival-based demand jumps. Her forecast models improve warehouse stocking efficiency by 23%, saving costs and delighting customers.

## Why Forecasting Matters

- **Business Planning:** Accurate forecasts help businesses manage inventory, staffing, budgeting, and expansion.
- **Healthcare:** Predicting patient inflow enables hospitals to prepare staff and resources accordingly.
- **Finance:** Stock market trends and economic indicators are forecasted to support investment decisions.
- **Energy:** Load forecasting helps optimize power grid distribution and renewable resource planning.
- **Climate Science:** Long-term forecasting of rainfall, temperature, or floods aids in disaster preparedness.

## Forecasting is Not Just Guesswork

Good forecasting is not about guessing — it's about learning from patterns. With time series models, we can:

- Identify repeating cycles (seasonality)
- Detect long-term movements (trend)
- Account for sudden changes (irregularities)
- Incorporate domain knowledge (e.g., holidays, promotions)

## Applications of Forecasting Across Domains

| Domain     | Forecasting Use Case                                                |
|------------|---------------------------------------------------------------------|
| Retail     | Predicting product demand, planning inventory, optimizing logistics |
| Healthcare | Forecasting patient inflow, ICU occupancy, disease outbreaks        |
| Finance    | Predicting stock prices, interest rates, credit risk                |
| Energy     | Forecasting electricity consumption, solar/wind output, grid load   |
| Climate    | Rainfall prediction, drought detection, temperature anomalies       |

Table 1.1: Key Domains Where Forecasting is Essential

### Practical Note for Students

In the later modules, we'll learn how to build forecasting models like Moving Averages, Exponential Smoothing, ARIMA, and even Deep Learning techniques. But before that, we must understand how to explore and decompose the time series — which we'll cover next.

## 1.3 Univariate vs Multivariate Time Series

### Story: Raju's Forecasting Dilemma at Airtel

**Raju**, a data scientist at **Airtel**, is assigned to forecast the number of mobile recharges per week for southern India. At first, he looks at historical recharge data—classic univariate time series. But his manager asks, "Can we also account for ad spend, competitor pricing, and national holidays?" Now Raju is entering multivariate territory. Understanding which model to choose is the first step in his data science journey.

Time series can be broadly categorized into two types:

- **Univariate Time Series:** A single variable recorded over consistent time intervals.
- **Multivariate Time Series:** Two or more related variables measured over time, potentially influencing one another.

### Univariate Time Series

A **univariate time series** involves a single variable tracked over time. It focuses solely on the internal structure of its values — trends, seasonality, and irregularities.

**Examples:**



- **Electric Utility Load Forecasting:** Predicting hourly electricity consumption based only on past usage data.
- **ATM Cash Withdrawals:** Modeling cash needs at a specific ATM using historical daily withdrawal records.
- **Online Learning Portal:** Estimating daily user logins based solely on past login counts.

#### Industry Insight

Univariate models are useful when external drivers are unavailable, or when we need quick and interpretable forecasts using tools like ARIMA, Holt-Winters, or Prophet.

## Multivariate Time Series

A **multivariate time series** consists of several time-dependent variables. These models capture the relationships and potential causality between variables.

#### Examples:

- **Telecom Data Usage Forecasting:** Predicting daily mobile data consumption using past consumption, current data plan, app download trends, and regional promotions.
- **Hospital ER Forecasting:** Modeling daily emergency room visits using past visits, pollution index, weather data, and public event schedules.
- **Stock Market Modeling:** Forecasting stock prices using indicators like volume, exchange rate, interest rates, and sector performance.
- **Smart City Traffic Forecasting:** Predicting traffic congestion based on past congestion levels, weather, road closures, and public events.
- **E-commerce Demand Forecasting:** Estimating product sales using features such as price, marketing campaign strength, holidays, and competitor activity.

#### Industry Insight

Multivariate forecasting is vital when the target variable depends on other factors. Techniques like VAR, VARMA, LSTM, and TCN are used for advanced multivariate modeling.

Visual Illustration

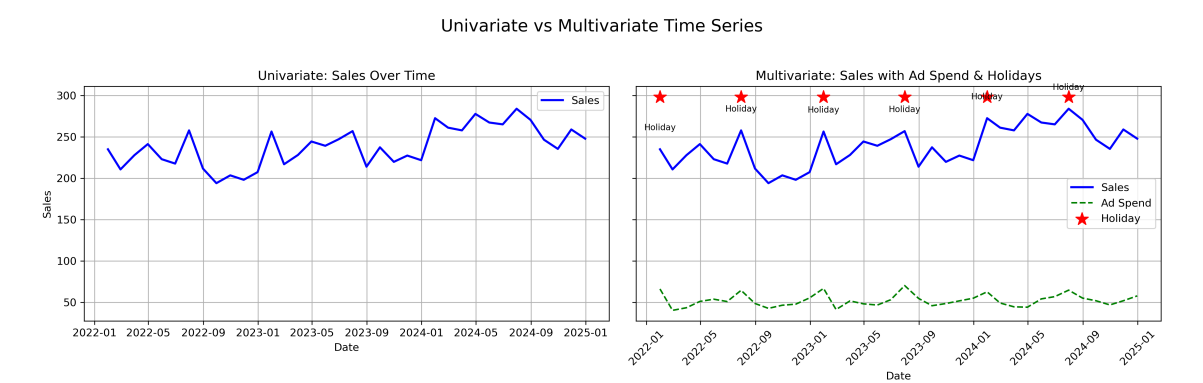


Figure 1.2: Visual Comparison: Univariate vs. Multivariate Time Series

Comparison Table

| Univariate Time Series                     | Multivariate Time Series                  |
|--------------------------------------------|-------------------------------------------|
| One variable tracked over time             | Two or more interrelated variables        |
| Captures internal dynamics only            | Considers influence of external variables |
| Simpler to build, interpret, and maintain  | More complex, often more accurate         |
| Used in operational or quick analysis      | Used in strategic, holistic forecasting   |
| Common Models: AR, MA, ARIMA, Holt-Winters | Common Models: VAR, VARMA, LSTM, TCN      |

Table 1.2: Comparison between Univariate and Multivariate Time Series

In Practice

When data is scarce or speed is critical, univariate models work well. For richer insights and higher accuracy, especially in competitive environments, multivariate models provide greater explanatory power. Always consider trade-offs like data availability, model complexity, and interpretability.

## 1.4 Special Features of Time Series

### Story: Swathi's Energy Forecasting Mission

Swathi is a data scientist at **Tata Power**, responsible for predicting hourly electricity demand. Her manager reminds her: "Energy usage doesn't just follow a trend — it spikes in summer, dips at night, and goes haywire during festivals!" Swathi soon realizes that time series data has hidden layers of complexity—seasonality, autocorrelation, and sudden shocks. Understanding these characteristics is vital before choosing any forecasting model.

Time series data exhibits unique statistical behaviors that differentiate it from random or cross-sectional data. Recognizing these features allows data scientists to apply appropriate transformations, models, and diagnostics.

### Key Features of Time Series Data

- **Trend:** A long-term increase or decrease in the series over time. For example, internet usage shows an upward trend globally.
- **Seasonality:** Regular, periodic fluctuations occurring within specific time intervals (e.g., daily, weekly, monthly). Retail sales typically spike during festive seasons.
- **Cyclic Behavior:** Long-term oscillations without fixed frequency, often driven by economic or market cycles. These are different from seasonality and are harder to model.
- **Autocorrelation:** The degree to which current values of a series relate to its past values. Strong autocorrelation means the past can meaningfully inform the future.
- **Irregularity (Noise):** Unpredictable fluctuations due to unforeseen events like natural disasters, policy changes, or market crashes.
- **Non-stationarity:** When statistical properties like mean and variance change over time. Most real-world time series are non-stationary and need to be transformed for effective modeling.

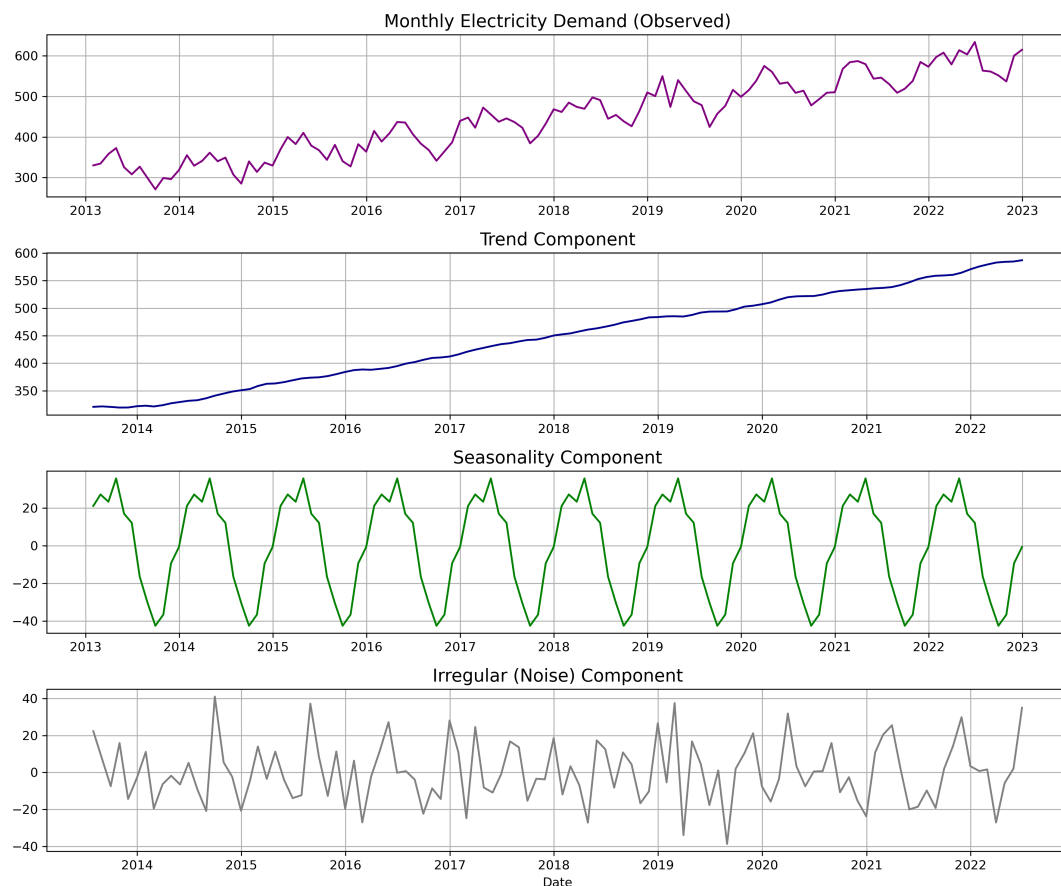


Figure 1.3: Decomposition of Monthly Electricity Demand into Trend, Seasonality, and Irregular Components

#### Did You Know?

The **Augmented Dickey-Fuller (ADF)** test is commonly used to check if a time series is stationary. It's a foundational step before applying ARIMA or related models.

#### Industry Tip

Don't jump into modeling before understanding the nature of your data. Visualize your time series, identify repeating patterns, check for stationarity, and examine the auto-correlation structure. These insights guide your preprocessing and model selection.

## 1.5 Exploratory Data Analysis (EDA) of Time Series

### Story: Raju Explores Recharge Patterns

After his manager at **Airtel** greenlights the forecasting project, Raju begins by diving into the raw time series data. He knows that before applying any models, understanding the structure and peculiarities of the data is essential. Is there a trend? Are there seasonal spikes during festival periods? Are there outliers after certain ad campaigns? These insights will shape every decision he makes in modeling.

### Why EDA Matters in Time Series

Exploratory Data Analysis (EDA) in time series isn't just about statistics—it's about storytelling with time. It helps us:

- Understand temporal patterns like trend, seasonality, and irregularities.
- Identify missing values, outliers, or data anomalies.
- Gain intuition on stationarity, periodicity, and volatility.
- Visually inspect changes in structure over time (shifts, breaks, etc.).

### Key EDA Techniques for Time Series

1. **Line Plots:** Visualize raw values over time.
2. **Rolling Statistics:** Compute rolling mean and standard deviation.
3. **Seasonal Plots:** Identify recurring patterns at regular intervals.
4. **Lag Plots:** Explore autocorrelation visually.
5. **Histogram and KDE:** Analyze distributional properties.
6. **Boxplots by Time Units:** Monthly, weekly boxplots to spot seasonality.

### Python Code: Performing EDA on a Time Series

Listing 1.1: Basic Time Series EDA in Python

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load example dataset: daily electricity demand (simulated)
df = pd.readcsv('electricitydemand.csv', parsedates=['Date'], indexcol='Date'
)

# Line plot
df['Demand'].plot(title='Electricity Demand Over Time', figsize=(12, 4))
```

```
plt.ylabel("MW")
plt.show()

# Rolling statistics
rollingmean = df['Demand'].rolling(window=30).mean()
rollingstd = df['Demand'].rolling(window=30).std()

plt.figure(figsize=(12, 4))
plt.plot(df['Demand'], label='Original')
plt.plot(rollingmean, label='Rolling Mean', color='red')
plt.plot(rollingstd, label='Rolling Std Dev', color='green')
plt.legend()
plt.title('Rolling Mean and Std Dev (30-day)')
plt.show()

# Seasonal plot
df['Month'] = df.index.month
sns.boxplot(x='Month', y='Demand', data=df)
plt.title('Monthly Seasonality Boxplot')
plt.show()
```

#### Visual Illustration Placeholder

- **Figure 1:** Line Plot of Electricity Demand
- **Figure 2:** Rolling Mean and Std Dev
- **Figure 3:** Boxplot of Monthly Demand Seasonality

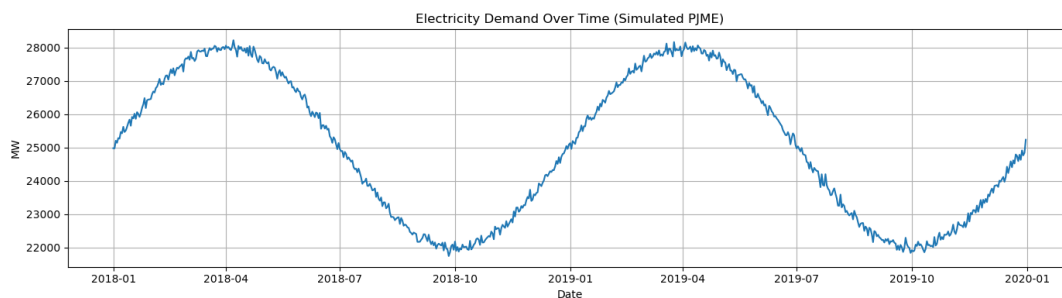
*[Instructor Note: Include annotated images of the above plots below this section for maximum engagement.]*

## Real Industry EDA Use Cases

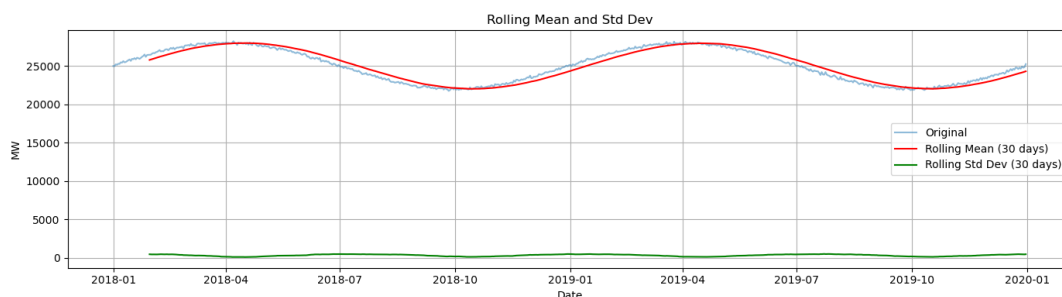
- **Banking:** Analyzing ATM withdrawals to adjust cash loading cycles.
- **Retail:** Understanding hourly footfall before running promotions.
- **Healthcare:** Identifying peaks in ER admissions due to seasonal illness.
- **Utilities:** Studying electricity demand before and after heatwaves.

#### In Practice

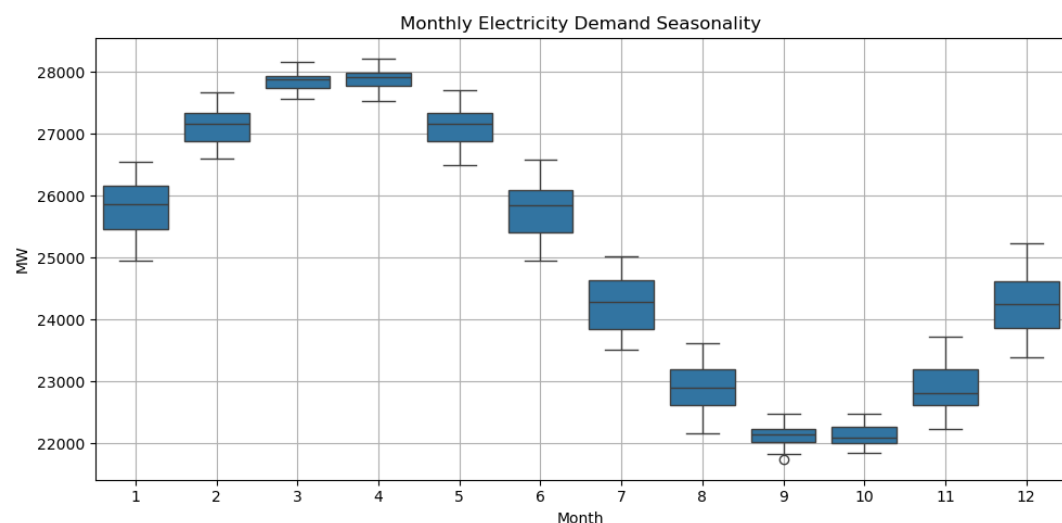
Great forecasters are first great explorers. Before using any ARIMA or LSTM model, always spend time inspecting, plotting, and questioning the data. The patterns you discover in EDA will often guide the modeling process more than any textbook formula.



**Figure:** Line Plot of Electricity Demand



**Figure:** Rolling Mean and Standard Deviation of Demand



**Figure:** Boxplot Showing Monthly Demand Seasonality

## 1.6 Components of a Time Series

### Story: Kavya's Energy Forecasting Assignment

Kavya, a data scientist at the State Energy Authority, is asked to forecast weekly electricity demand. Her supervisor mentions, “Make sure your model accounts for summer spikes and weekly dips.” Kavya realizes that understanding the underlying components of the time series is critical before choosing a forecasting model. She dives into decomposition to uncover the hidden signals in the data.

## Why Decompose Time Series?

A time series is not just a stream of values — it's a blend of distinct patterns and randomness. Decomposing a time series into its components helps:

- Understand the structure of data
- Choose appropriate forecasting models
- Detect anomalies or structural shifts
- Communicate findings effectively

## Key Components of Time Series

1. **Trend ( $T_t$ ):** The long-term progression of the series. It reflects persistent upward or downward movements.
2. **Seasonality ( $S_t$ ):** Repeating short-term cycles at fixed intervals (e.g., daily, weekly, yearly).
3. **Irregular or Residual ( $R_t$ ):** Random noise or events that cannot be attributed to trend or seasonality.

## Mathematical Models

### Additive Model:

$$Y_t = T_t + S_t + R_t$$

### Multiplicative Model:

$$Y_t = T_t \times S_t \times R_t$$

#### When to Use Additive vs Multiplicative?

Use the **additive model** when seasonal fluctuations are roughly constant. Use the **multiplicative model** when seasonal effects scale with the level of the series.



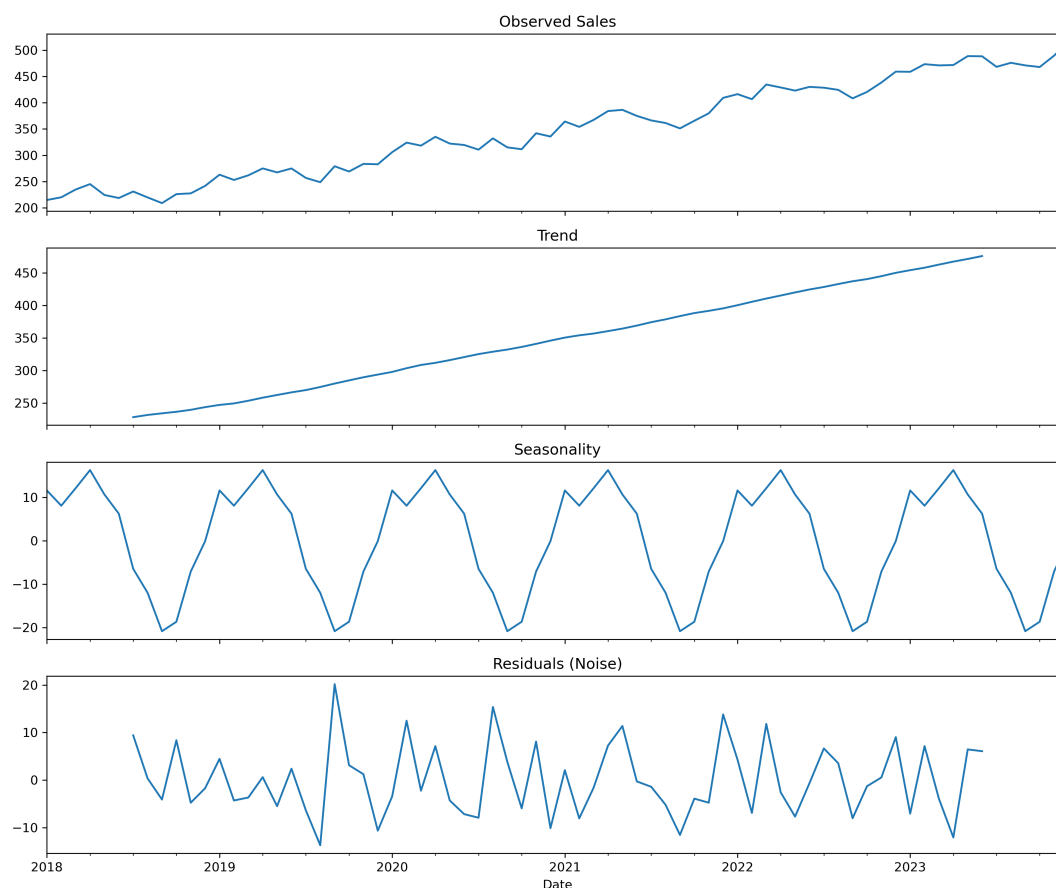


Figure 1.4: Decomposition of Monthly Retail Sales Time Series into Trend, Seasonality, and Residuals

## Examples from Industry

- **Retail:** Weekly demand has a trend driven by population growth, seasonality from festivals, and random spikes from flash sales.
- **Weather Forecasting:** Daily temperature data shows a yearly sinusoidal seasonal pattern and long-term trends due to climate change.
- **Healthcare:** ER visit patterns show a weekday-weekend seasonality and long-term upward trend during flu seasons.
- **Energy:** Hourly electricity demand shows seasonal daily peaks (morning/evening) and long-term trends due to urbanization.

### Practical Insight

Decomposing time series before modeling helps avoid overfitting and leads to more explainable forecasts. Most modern algorithms benefit from this pre-analysis.

## 1.7 Time Series Decomposition: Additive vs Multiplicative Models

### Story: Rani's Market Demand Puzzle at BigBazaar

Rani, a forecasting lead at **BigBazaar**, noticed that during festivals, the sales of sugar increase dramatically. But was this a seasonal spike or just a random fluctuation? To understand this, she decomposed the time series into its underlying components—trend, seasonality, and noise—to isolate what's driving customer behavior.

Decomposition helps us understand **how a time series is built** by separating it into three main components:

- **Trend (T):** The long-term progression of the series.
- **Seasonality (S):** Repeating short-term patterns (e.g., yearly, monthly, weekly).
- **Irregular or Residual (R):** Random noise that cannot be explained by trend or seasonality.

There are two main models for decomposition:

### Additive Model

$$Y_t = T_t + S_t + R_t$$

Use this when:

- The magnitude of seasonal variation is roughly constant over time.
- The trend and seasonality are independent.

**Example:** Daily website traffic increases during weekends consistently by about 300 visits, irrespective of total traffic levels.

### Multiplicative Model

$$Y_t = T_t \times S_t \times R_t$$

Use this when:

- The seasonal fluctuations grow with the level of the time series.
- Seasonality is proportional to trend.

**Example:** Holiday sales in December grow every year, but the peak becomes taller as the overall business grows.

## Visual Comparison

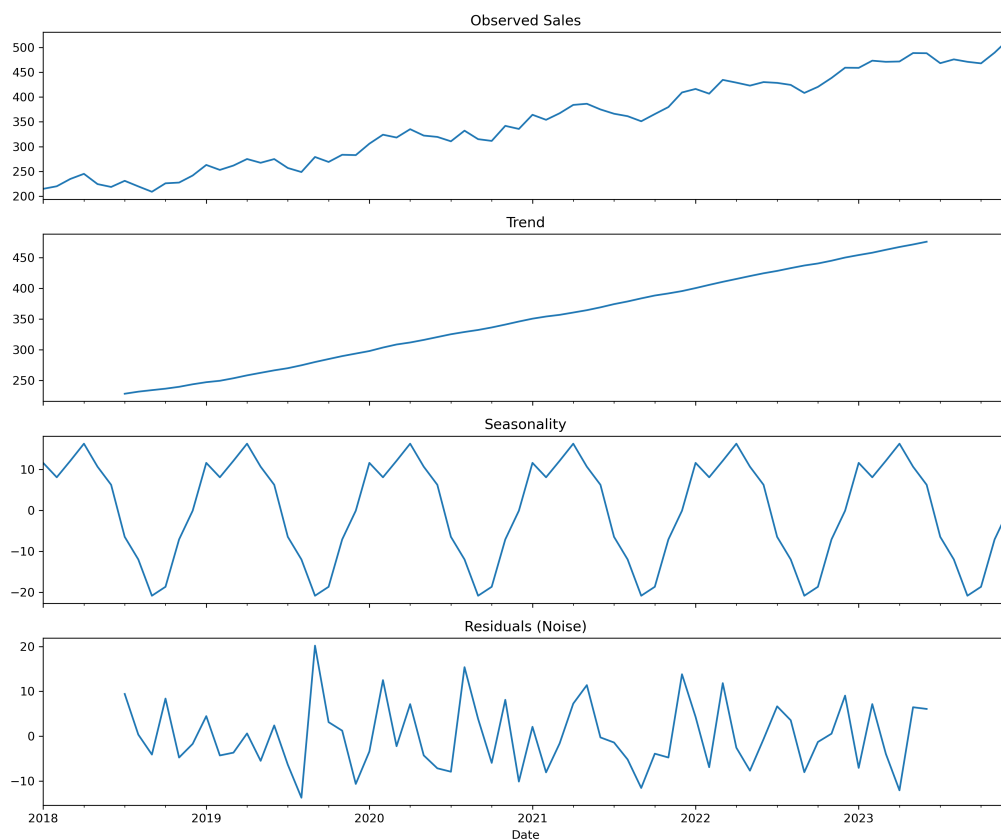


Figure 1.5: Additive Decomposition of Monthly Retail Sales Time Series

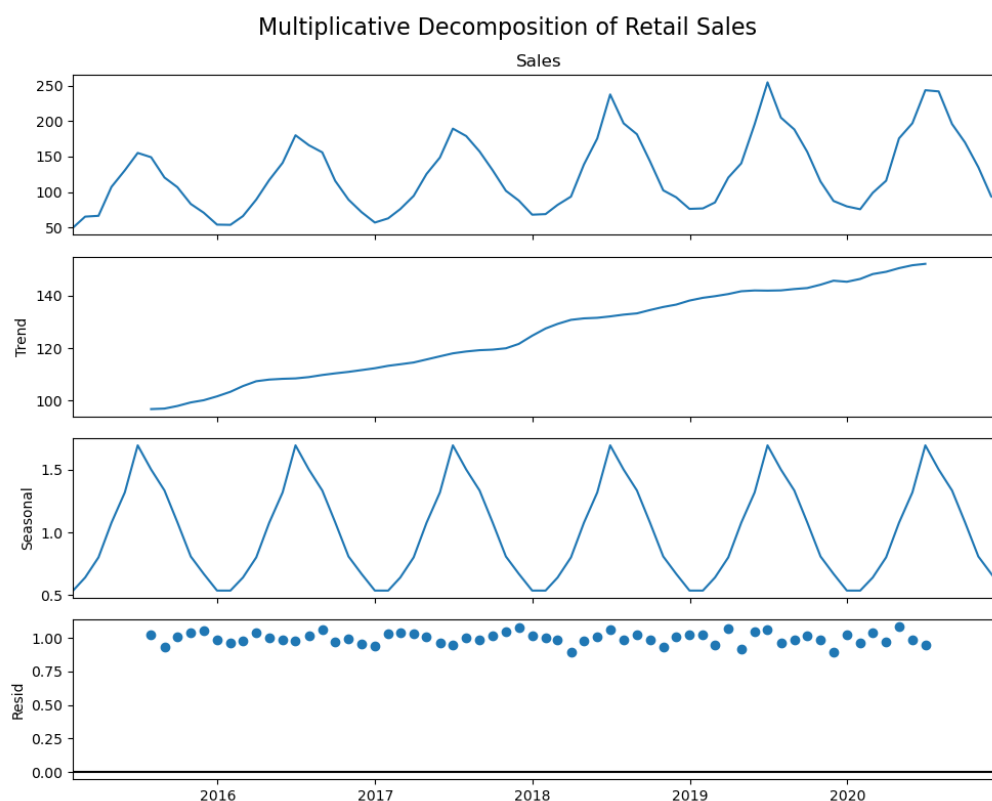


Figure 1.6: Multiplicative Decomposition of Monthly Retail Sales Time Series

### Pro Tip

If you're unsure whether to use additive or multiplicative decomposition, try plotting the time series. If seasonal variation looks constant  $\rightarrow$  additive. If it increases with time  $\rightarrow$  multiplicative.

### In Practice

Libraries like `statsmodels` provide ready-to-use tools to perform seasonal decomposition. In real-world forecasting pipelines, this is often the first step before model selection.

## 1.8 Visualizing Time Series Patterns

### Story: Meera's Morning Dashboard at BigMart

**Meera**, a senior analyst at **BigMart Retail**, starts every morning by reviewing sales trends across regions. She doesn't jump into statistical models — her first step is always visual. "Patterns reveal themselves best to the naked eye," she often says. With line charts, seasonal subseries plots, and autocorrelation diagrams, Meera can spot anomalies, growth spurts, and cyclical dips — all before running a single line of code.

Visualizing time series data is one of the most powerful and intuitive methods for identifying hidden patterns, detecting anomalies, and understanding seasonality, trends, and cyclic behavior. Before applying any statistical modeling, visualization helps build a strong mental model of the data.

## 1. Line Plots

A simple but powerful tool to observe:

- **Trend direction** (upward/downward)
- **Seasonal patterns** (repeating cycles)
- **Shocks or outliers** (sudden changes)

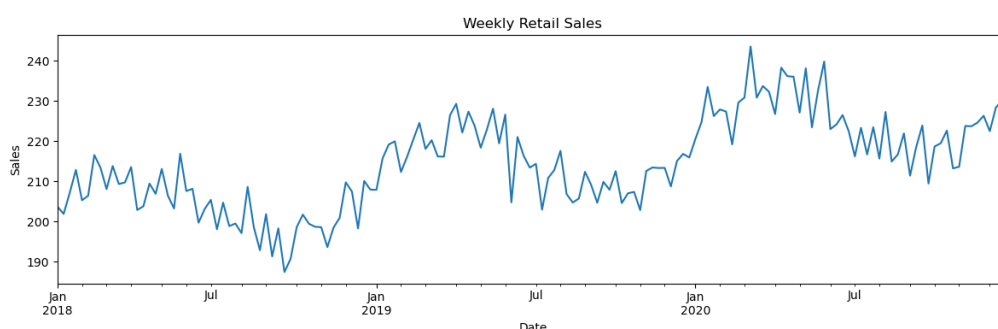


Figure 1.7: Line Plot of Weekly Retail Sales

## 2. Seasonal Subseries Plots

These plots display data broken down by time units (e.g., months or weeks) across years, helping visualize recurring patterns.

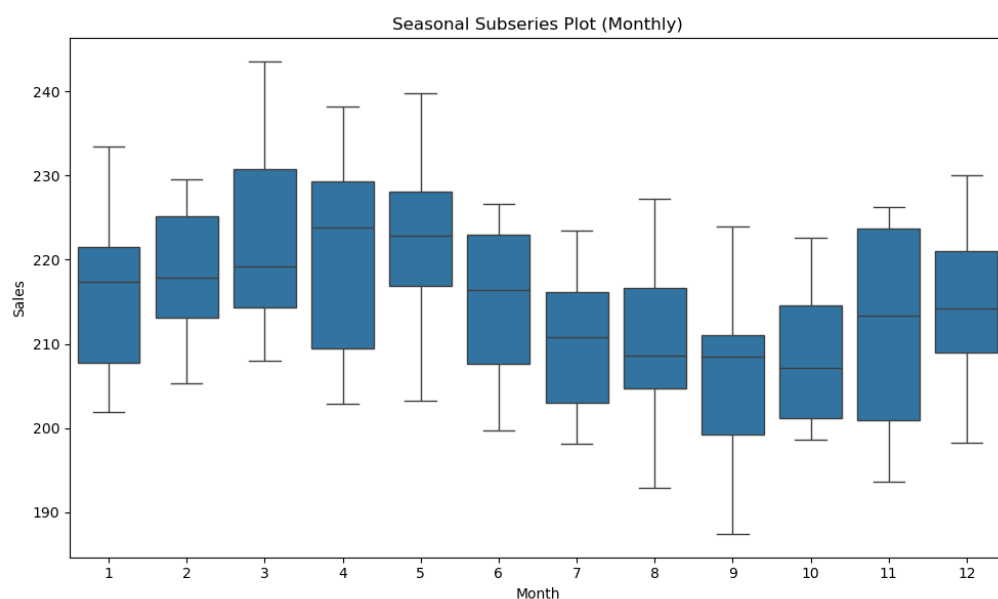


Figure 1.8: Seasonal Subseries Plot of Monthly Sales

### 3. Autocorrelation and Partial Autocorrelation Plots

These plots help examine internal dependencies within the series:

- **Autocorrelation Function (ACF):** Correlation of the time series with its own past values.
- **Partial Autocorrelation Function (PACF):** Correlation after removing intermediate lag effects.

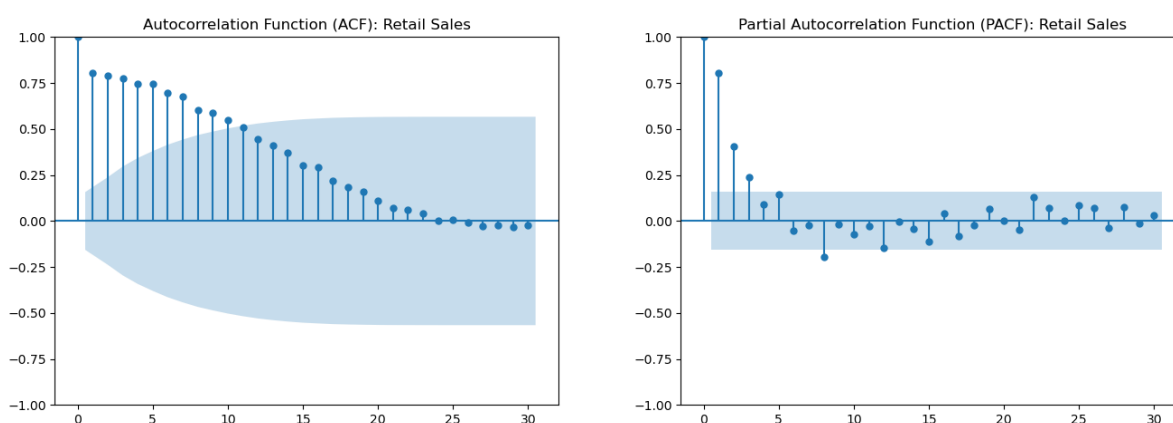


Figure 1.9: ACF and PACF for Retail Sales Time Series

#### In Practice

Autocorrelation plots are especially useful to select lags for ARIMA models or to assess the memory of a process. Visual patterns like sharp cutoff in PACF or gradual decay in ACF guide modeling decisions.

### 4. Seasonal Decomposition Plots

Visualization of decomposed components such as trend, seasonality, and residuals (from STL or classical decomposition) is key to understanding underlying drivers.

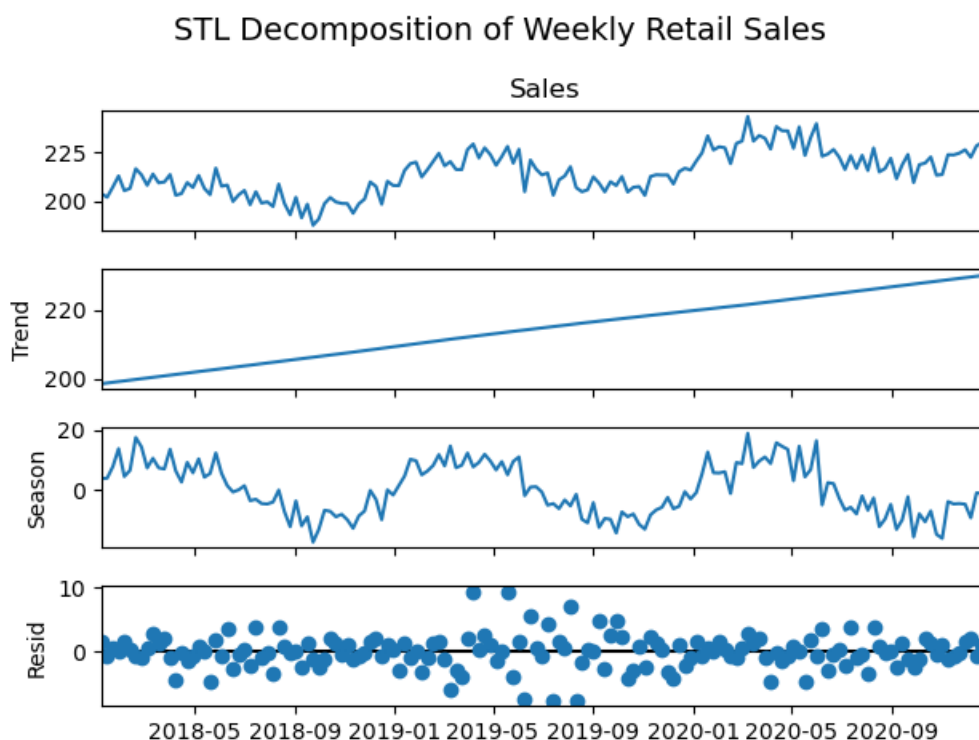


Figure 1.10: STL Decomposition of Weekly Retail Sales

## 5. Heatmaps and Calendar Plots (Optional)

These offer a high-level overview of daily/weekly values over longer periods, useful in identifying systematic daily or weekly variation.

### When to Use What

- Use **line plots** for high-level trend scanning.
- Use **seasonal plots** to spot regular patterns.
- Use **ACF/PACF** to inform statistical models.
- Use **decomposition plots** for model selection and diagnostics.
- Use **heatmaps/calendar plots** to visualize dense patterns over long periods.